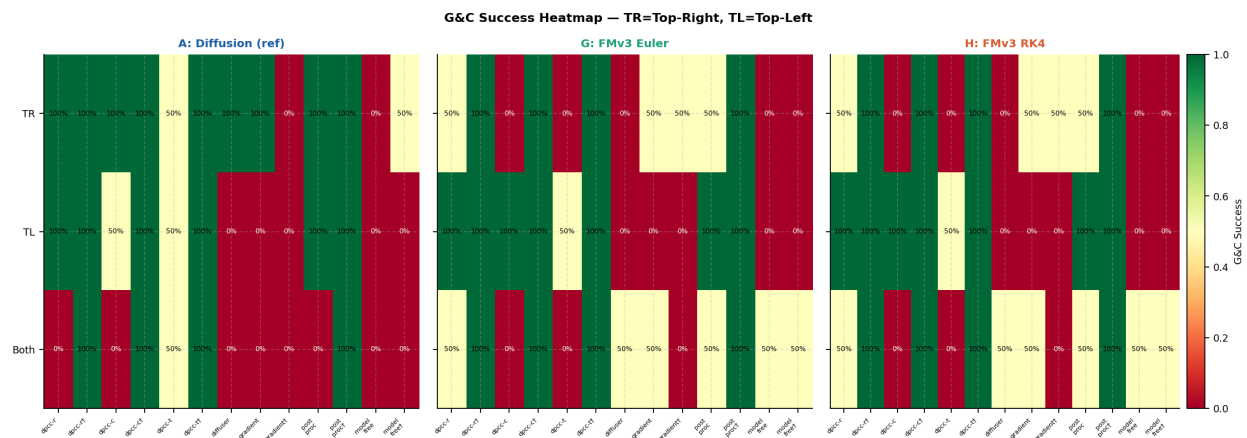


FMv3 RK4 Analysis: Euler vs RK4 ODE Solver

Three-way comparison — A: Diffusion S6 (ref) · G: FMv3 Euler · H: FMv3 RK4

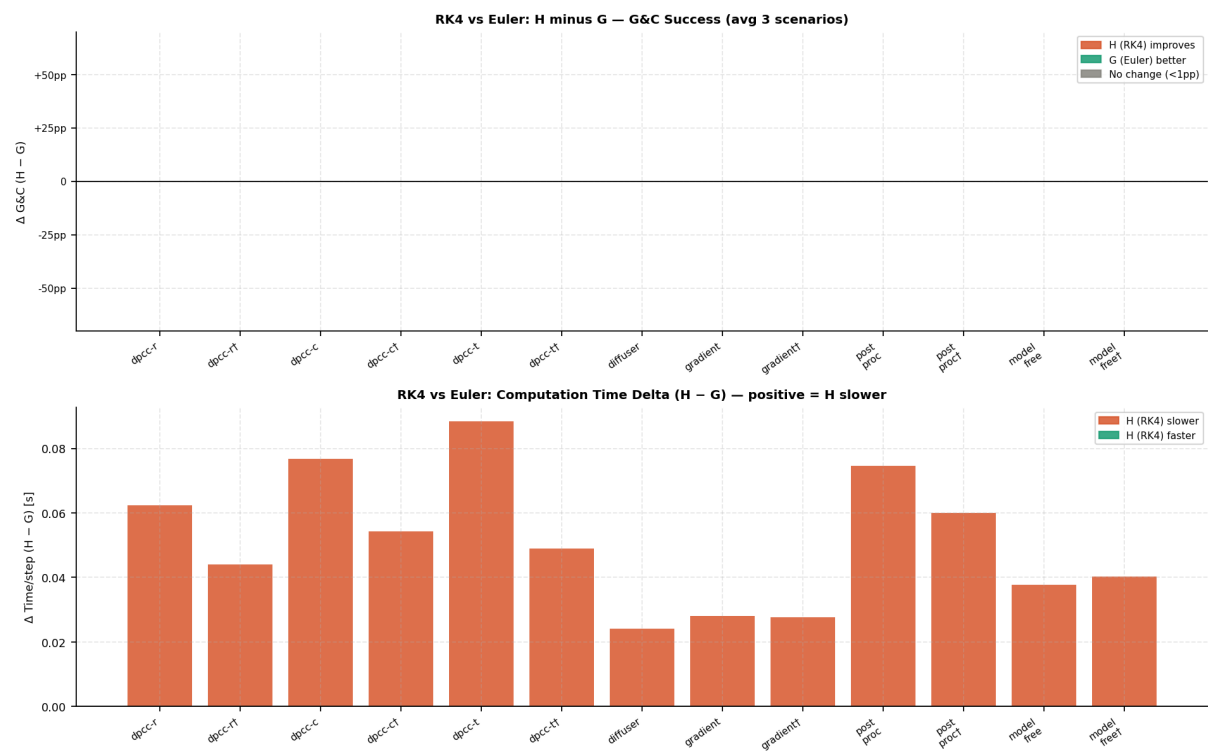
A = Diffusion S6 · reference baseline	G = FMv3 Euler · explicit Euler, ODE=10, continuous-t	H = FMv3 RK4 · Runge-Kutta 4th order, ODE=10, continuous-t
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Key change in H (FMv3 RK4): the ODE integration at each step uses **Runge-Kutta 4th order (RK4)** instead of the explicit Euler method used in G. RK4 performs 4 velocity-field evaluations per ODE step (vs 1 for Euler) giving higher-order integration accuracy, at the cost of roughly 4× the per-step inference budget. All other settings are identical: FMv3 architecture (continuous real-valued t), action_weight=1, ODE steps=10, checkpoint step 90000, Seed 6. Comparing H vs G isolates the pure effect of the ODE solver order.



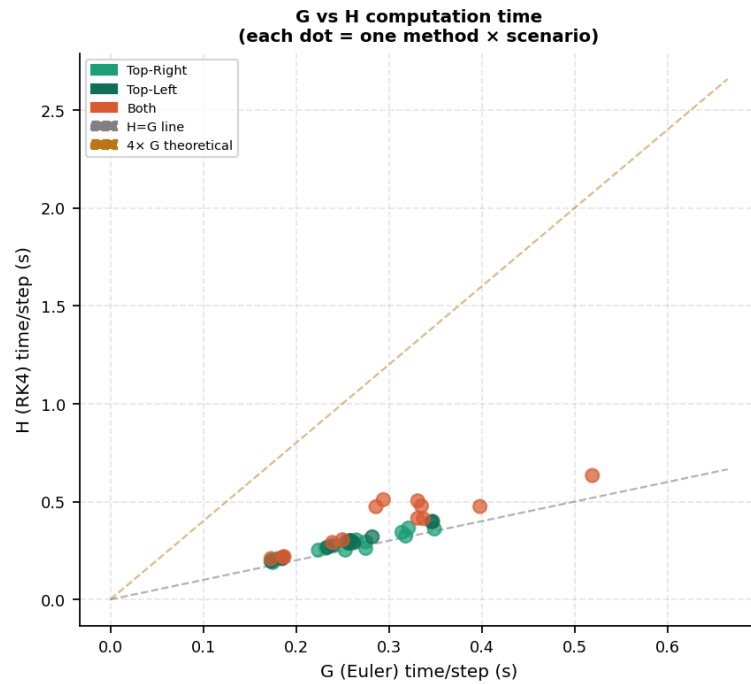
1.3 Euler → RK4 Impact: H minus G

Top panel: G&C; change from switching to RK4 (avg 3 scenarios). Coral = H (RK4) improves, teal = G (Euler) was better, gray = no change (<1pp). Bottom panel: computation time delta — positive = RK4 is slower.

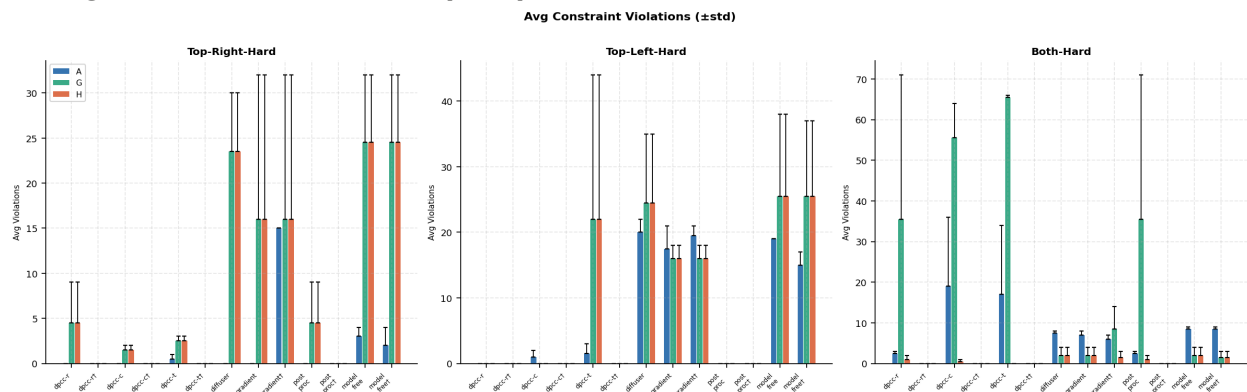


1.4 Computation Time: G (Euler) vs H (RK4) Scatter

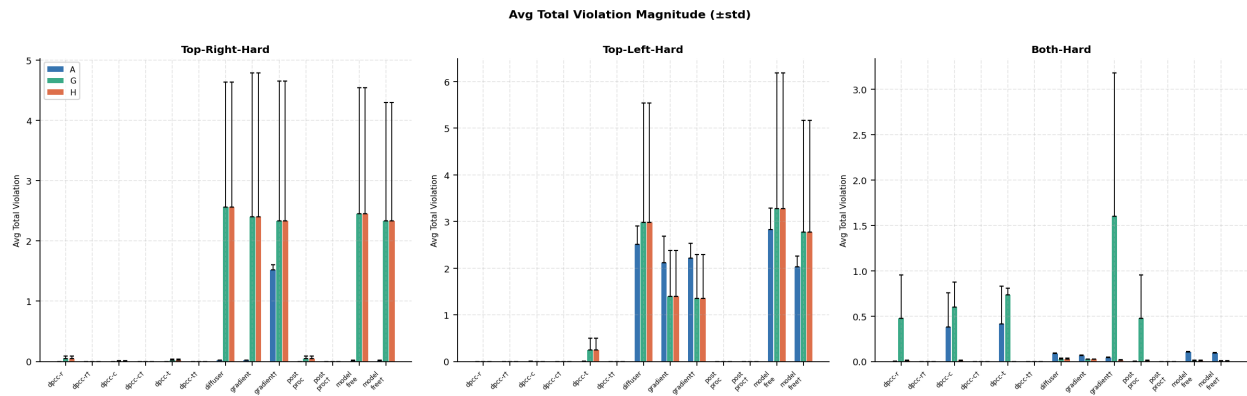
Each dot = one method $\times$ scenario. The dashed gray line is $H=G$ (no overhead). The amber dashed line is the theoretical $4\times$ ceiling for RK4 (4 function evals/step). Points far above the gray line indicate methods where RK4 overhead is significant.



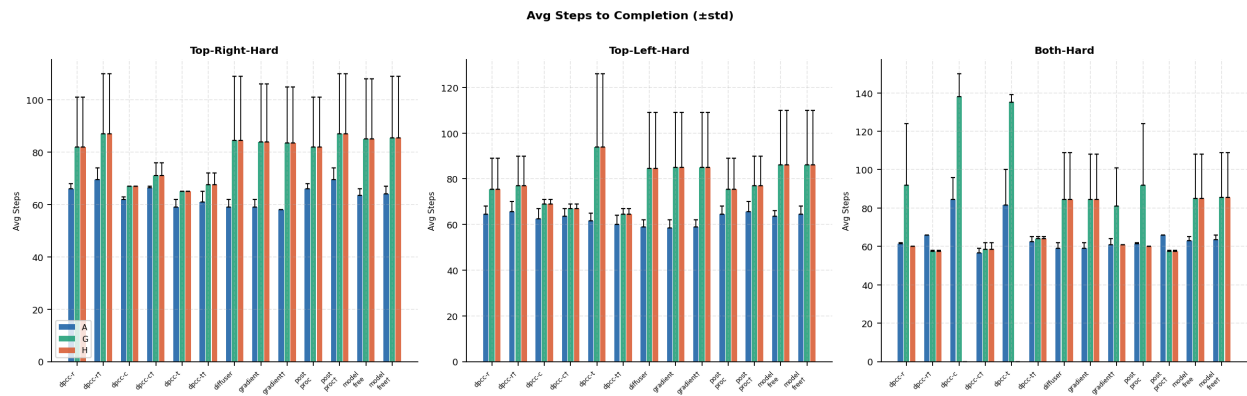
1.5 Avg Constraint Violations ($\pm$ std)



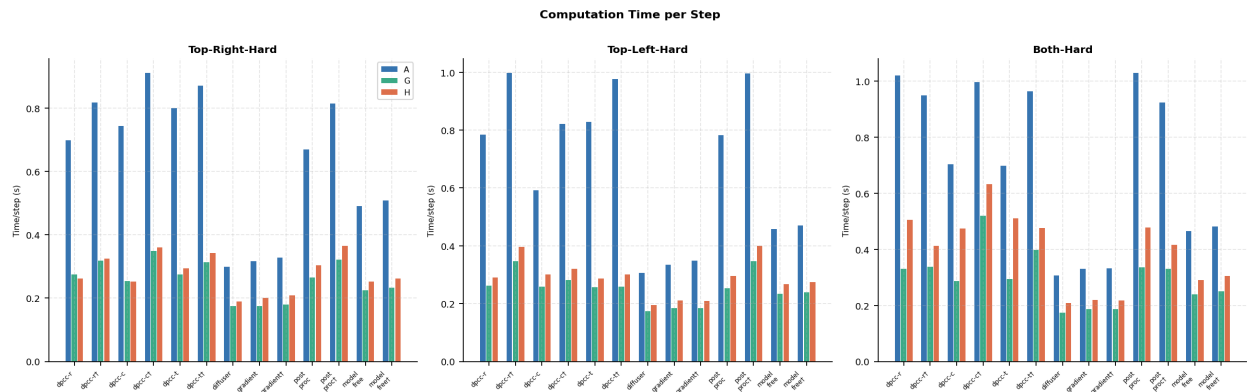
1.6 Avg Total Violation Magnitude ($\pm$ std)



1.7 Avg Steps to Completion ($\pm$ std)

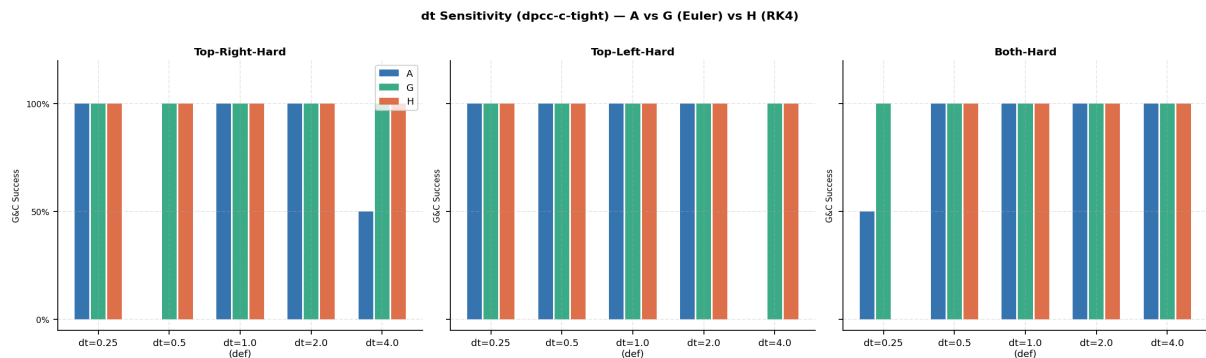


1.8 Computation Time per Step

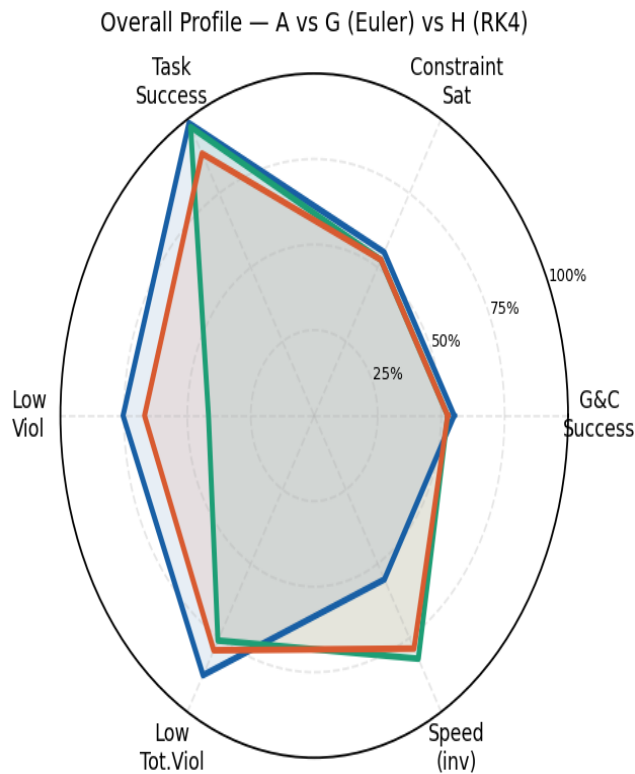


1.9 dt Sensitivity — dpcc-c-tightened: A vs G vs H

G&C; success across $dt \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$. H reintroduces the $dt=0.25$ failure on both-hard (0%) that G had solved. Top-left and top-right dt sensitivity is identical between G and H.



1.10 Overall Model Profile Radar



Section 2 — Complete Raw Data Tables

All metrics for H (FMv3 RK4), G (FMv3 Euler), and A (Diffusion). **SR**=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps±std**, **Viol#±std**, **Viol Σ ±std**, **Time(s/step)**, **TE**=Tracking Error.

2.1 Top-Right-Hard

H: FMv3 RK4 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	82.0±19.0	4.5±4.5	0.045±0.045	0.261	–
dpcc-r-tight	100%	100%	100%	87.0±23.0	0.0±0.0	0.000±0.000	0.324	–
dpcc-c	50%	0%	0%	67.0±0.0	1.5±0.5	0.005±0.004	0.252	–
dpcc-c-tight	100%	100%	100%	71.0±5.0	0.0±0.0	0.000±0.000	0.360	–
dpcc-t	50%	0%	0%	65.0±0.0	2.5±0.5	0.018±0.016	0.294	–
dpcc-t-tight	100%	100%	100%	67.5±4.5	0.0±0.0	0.000±0.000	0.343	–
diffuser	100%	0%	0%	84.5±24.5	23.5±6.5	2.563±2.077	0.189	0.029
gradient	100%	50%	50%	84.0±22.0	16.0±16.0	2.405±2.386	0.201	–
gradient-tight	100%	50%	50%	83.5±21.5	16.0±16.0	2.337±2.318	0.208	–
post_proc	100%	50%	50%	82.0±19.0	4.5±4.5	0.045±0.045	0.304	–
post_proc-tight	100%	100%	100%	87.0±23.0	0.0±0.0	0.000±0.000	0.365	–
model_free	100%	0%	0%	85.0±23.0	24.5±7.5	2.453±2.092	0.252	–
model_free-tight	100%	0%	0%	85.5±23.5	24.5±7.5	2.334±1.967	0.262	–
dpcc-c-tight-dt0.25	100%	100%	100%	78.5±3.5	0.0±0.0	0.002±0.000	0.769	–
dpcc-c-tight-dt0.5	100%	100%	100%	77.5±5.5	0.0±0.0	0.001±0.000	0.489	–
dpcc-c-tight-dt2.0	100%	100%	100%	71.0±6.0	0.0±0.0	0.000±0.000	0.330	–
dpcc-c-tight-dt4.0	100%	100%	100%	131.5±1.5	0.0±0.0	0.000±0.000	0.290	–

G: FMv3 Euler — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	82.0±19.0	4.5±4.5	0.045±0.045	0.275	–
dpcc-r-tight	100%	100%	100%	87.0±23.0	0.0±0.0	0.000±0.000	0.318	–
dpcc-c	50%	0%	0%	67.0±0.0	1.5±0.5	0.005±0.004	0.253	–
dpcc-c-tight	100%	100%	100%	71.0±5.0	0.0±0.0	0.000±0.000	0.349	–
dpcc-t	50%	0%	0%	65.0±0.0	2.5±0.5	0.018±0.016	0.275	–
dpcc-t-tight	100%	100%	100%	67.5±4.5	0.0±0.0	0.000±0.000	0.314	–
diffuser	100%	0%	0%	84.5±24.5	23.5±6.5	2.563±2.077	0.175	0.029
gradient	100%	50%	50%	84.0±22.0	16.0±16.0	2.405±2.386	0.175	–
gradient-tight	100%	50%	50%	83.5±21.5	16.0±16.0	2.337±2.318	0.180	–
post_proc	100%	50%	50%	82.0±19.0	4.5±4.5	0.045±0.045	0.265	–
post_proc-tight	100%	100%	100%	87.0±23.0	0.0±0.0	0.000±0.000	0.321	–
model_free	100%	0%	0%	85.0±23.0	24.5±7.5	2.453±2.092	0.224	–
model_free-tight	100%	0%	0%	85.5±23.5	24.5±7.5	2.334±1.967	0.232	–
dpcc-c-tight-dt0.25	100%	100%	100%	78.5±3.5	0.0±0.0	0.002±0.000	0.681	–

dpcc-c-tight-dt0.5	100%	100%	100%	77.5±5.5	0.0±0.0	0.001±0.000	0.426	–
dpcc-c-tight-dt2.0	100%	100%	100%	71.0±6.0	0.0±0.0	0.000±0.000	0.272	–
dpcc-c-tight-dt4.0	100%	100%	100%	131.5±1.5	0.0±0.0	0.000±0.000	0.270	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.698	–
dpcc-r-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.817	–
dpcc-c	100%	100%	100%	62.0±1.0	0.0±0.0	0.000±0.000	0.744	–
dpcc-c-tight	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.911	–
dpcc-t	100%	50%	50%	59.0±3.0	0.5±0.5	0.000±0.000	0.799	–
dpcc-t-tight	100%	100%	100%	61.0±4.0	0.0±0.0	0.000±0.000	0.871	–
diffuser	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.298	0.025
gradient	100%	100%	100%	59.0±3.0	0.0±0.0	0.021±0.002	0.316	–
gradient-tight	50%	0%	0%	58.0±0.0	15.0±0.0	1.521±0.082	0.327	–
post_proc	100%	100%	100%	66.0±2.0	0.0±0.0	0.000±0.000	0.670	–
post_proc-tight	100%	100%	100%	69.5±4.5	0.0±0.0	0.000±0.000	0.814	–
model_free	100%	0%	0%	63.5±2.5	3.0±1.0	0.012±0.011	0.490	–
model_free-tight	100%	50%	50%	64.0±3.0	2.0±2.0	0.009±0.009	0.508	–
dpcc-c-tight-dt0.25	100%	100%	100%	81.0±6.0	0.0±0.0	0.002±0.000	2.225	–
dpcc-c-tight-dt0.5	100%	0%	0%	75.5±2.5	1.0±0.0	0.003±0.000	1.668	–
dpcc-c-tight-dt2.0	100%	100%	100%	66.5±0.5	0.0±0.0	0.000±0.000	0.689	–
dpcc-c-tight-dt4.0	50%	50%	50%	72.0±0.0	0.0±0.0	0.000±0.000	0.535	–

2.2 Top-Left-Hard

H: FMv3 RK4 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	75.5±13.5	0.0±0.0	0.000±0.000	0.290	–
dpcc-r-tight	100%	100%	100%	77.0±13.0	0.0±0.0	0.000±0.000	0.396	–
dpcc-c	100%	100%	100%	69.0±2.0	0.0±0.0	0.000±0.000	0.301	–
dpcc-c-tight	100%	100%	100%	67.0±2.0	0.0±0.0	0.000±0.000	0.320	–
dpcc-t	100%	50%	50%	94.0±32.0	22.0±22.0	0.250±0.250	0.287	–
dpcc-t-tight	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.300	–
diffuser	100%	0%	0%	84.5±24.5	24.5±10.5	2.978±2.559	0.195	0.029
gradient	100%	0%	0%	85.0±24.0	16.0±2.0	1.391±0.993	0.211	–
gradient-tight	100%	0%	0%	85.0±24.0	16.0±2.0	1.351±0.943	0.209	–
post_proc	100%	100%	100%	75.5±13.5	0.0±0.0	0.000±0.000	0.296	–
post_proc-tight	100%	100%	100%	77.0±13.0	0.0±0.0	0.000±0.000	0.399	–
model_free	100%	0%	0%	86.0±24.0	25.5±12.5	3.278±2.909	0.267	–
model_free-tight	100%	0%	0%	86.0±24.0	25.5±11.5	2.773±2.399	0.275	–
dpcc-c-tight-dt0.25	100%	100%	100%	76.0±2.0	0.0±0.0	0.000±0.000	0.594	–

dpcc-c-tight-dt0.5	100%	100%	100%	68.5±2.5	0.0±0.0	0.000±0.000	0.366	–
dpcc-c-tight-dt2.0	100%	100%	100%	67.0±3.0	0.0±0.0	0.000±0.000	0.287	–
dpcc-c-tight-dt4.0	100%	100%	100%	124.5±4.5	0.0±0.0	0.000±0.000	0.285	–

G: FMV3 Euler — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	75.5±13.5	0.0±0.0	0.000±0.000	0.262	–
dpcc-r-tight	100%	100%	100%	77.0±13.0	0.0±0.0	0.000±0.000	0.346	–
dpcc-c	100%	100%	100%	69.0±2.0	0.0±0.0	0.000±0.000	0.258	–
dpcc-c-tight	100%	100%	100%	67.0±2.0	0.0±0.0	0.000±0.000	0.282	–
dpcc-t	100%	50%	50%	94.0±32.0	22.0±22.0	0.250±0.250	0.257	–
dpcc-t-tight	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.259	–
diffuser	100%	0%	0%	84.5±24.5	24.5±10.5	2.978±2.559	0.173	0.029
gradient	100%	0%	0%	85.0±24.0	16.0±2.0	1.391±0.993	0.185	–
gradient-tight	100%	0%	0%	85.0±24.0	16.0±2.0	1.351±0.943	0.185	–
post_proc	100%	100%	100%	75.5±13.5	0.0±0.0	0.000±0.000	0.254	–
post_proc-tight	100%	100%	100%	77.0±13.0	0.0±0.0	0.000±0.000	0.347	–
model_free	100%	0%	0%	86.0±24.0	25.5±12.5	3.278±2.909	0.234	–
model_free-tight	100%	0%	0%	86.0±24.0	25.5±11.5	2.773±2.399	0.239	–
dpcc-c-tight-dt0.25	100%	100%	100%	76.0±2.0	0.0±0.0	0.000±0.000	0.529	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±2.5	0.0±0.0	0.000±0.000	0.329	–
dpcc-c-tight-dt2.0	100%	100%	100%	67.0±3.0	0.0±0.0	0.000±0.000	0.268	–
dpcc-c-tight-dt4.0	100%	100%	100%	124.5±4.5	0.0±0.0	0.000±0.000	0.254	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.784	–
dpcc-r-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.998	–
dpcc-c	100%	50%	50%	62.5±4.5	1.0±1.0	0.003±0.003	0.592	–
dpcc-c-tight	100%	100%	100%	63.5±3.5	0.0±0.0	0.000±0.000	0.822	–
dpcc-t	100%	50%	50%	61.5±3.5	1.5±1.5	0.004±0.004	0.829	–
dpcc-t-tight	100%	100%	100%	60.0±4.0	0.0±0.0	0.000±0.000	0.976	–
diffuser	100%	0%	0%	59.0±3.0	20.0±2.0	2.515±0.387	0.306	0.025
gradient	100%	0%	0%	58.5±3.5	17.5±3.5	2.116±0.571	0.334	–
gradient-tight	100%	0%	0%	59.0±3.0	19.5±1.5	2.218±0.316	0.348	–
post_proc	100%	100%	100%	64.5±3.5	0.0±0.0	0.000±0.000	0.782	–
post_proc-tight	100%	100%	100%	65.5±4.5	0.0±0.0	0.000±0.000	0.996	–
model_free	100%	0%	0%	63.5±2.5	19.0±0.0	2.829±0.455	0.458	–
model_free-tight	100%	0%	0%	64.5±3.5	15.0±2.0	2.029±0.233	0.470	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0±3.0	0.0±0.0	0.000±0.000	1.259	–
dpcc-c-tight-dt0.5	100%	100%	100%	63.5±0.5	0.0±0.0	0.000±0.000	0.937	–

dpcc-c-tight-dt2.0	100%	100%	100%	64.5±0.5	0.0±0.0	0.000±0.000	0.679	–
dpcc-c-tight-dt4.0	100%	0%	0%	107.0±2.0	16.5±0.5	3.711±0.053	0.810	–

2.3 Both-Hard

H: FMv3 RK4 — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	50%	50%	50%	60.0±0.0	1.0±1.0	0.007±0.007	0.504	–
dpcc-r-tight	100%	100%	100%	57.5±0.5	0.0±0.0	0.000±0.000	0.413	–
dpcc-c	0%	0%	0%	0.0±0.0	0.5±0.5	0.006±0.006	0.474	–
dpcc-c-tight	100%	100%	100%	58.5±3.5	0.0±0.0	0.000±0.000	0.633	–
dpcc-t	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.510	–
dpcc-t-tight	100%	100%	100%	64.0±1.0	0.0±0.0	0.000±0.000	0.475	–
diffuser	100%	50%	50%	84.5±24.5	2.0±2.0	0.026±0.010	0.209	0.029
gradient	100%	50%	50%	84.5±23.5	2.0±2.0	0.023±0.005	0.219	–
gradient-tight	50%	0%	0%	61.0±0.0	1.5±1.5	0.015±0.006	0.217	–
post_proc	50%	50%	50%	60.0±0.0	1.0±1.0	0.007±0.007	0.478	–
post_proc-tight	100%	100%	100%	57.5±0.5	0.0±0.0	0.000±0.000	0.415	–
model_free	100%	50%	50%	85.0±23.0	2.0±2.0	0.006±0.006	0.291	–
model_free-tight	100%	50%	50%	85.5±23.5	1.5±1.5	0.003±0.003	0.305	–
dpcc-c-tight-dt0.25	0%	0%	0%	0.0±0.0	0.0±0.0	0.000±0.000	0.969	–
dpcc-c-tight-dt0.5	100%	100%	100%	62.0±5.0	0.0±0.0	0.001±0.001	0.554	–
dpcc-c-tight-dt2.0	100%	100%	100%	72.5±8.5	0.0±0.0	0.000±0.000	0.401	–
dpcc-c-tight-dt4.0	100%	100%	100%	101.0±2.0	0.0±0.0	0.000±0.000	0.383	–

G: FMv3 Euler — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	50%	50%	92.0±32.0	35.5±35.5	0.478±0.478	0.331	–
dpcc-r-tight	100%	100%	100%	57.5±0.5	0.0±0.0	0.000±0.000	0.337	–
dpcc-c	100%	0%	0%	138.0±12.0	55.5±8.5	0.598±0.276	0.286	–
dpcc-c-tight	100%	100%	100%	58.5±3.5	0.0±0.0	0.000±0.000	0.519	–
dpcc-t	100%	0%	0%	135.0±4.0	65.5±0.5	0.736±0.074	0.294	–
dpcc-t-tight	100%	100%	100%	64.0±1.0	0.0±0.0	0.000±0.000	0.398	–
diffuser	100%	50%	50%	84.5±24.5	2.0±2.0	0.026±0.010	0.173	0.029
gradient	100%	50%	50%	84.5±23.5	2.0±2.0	0.023±0.005	0.187	–
gradient-tight	100%	0%	0%	81.0±20.0	8.5±5.5	1.602±1.581	0.186	–
post_proc	100%	50%	50%	92.0±32.0	35.5±35.5	0.478±0.478	0.335	–
post_proc-tight	100%	100%	100%	57.5±0.5	0.0±0.0	0.000±0.000	0.331	–
model_free	100%	50%	50%	85.0±23.0	2.0±2.0	0.006±0.006	0.239	–
model_free-tight	100%	50%	50%	85.5±23.5	1.5±1.5	0.003±0.003	0.250	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0±2.0	0.0±0.0	0.000±0.000	0.666	–
dpcc-c-tight-dt0.5	100%	100%	100%	62.0±5.0	0.0±0.0	0.001±0.001	0.449	–

dpcc-c-tight-dt2.0	100%	100%	100%	72.5±8.5	0.0±0.0	0.000±0.000	0.323	–
dpcc-c-tight-dt4.0	100%	100%	100%	101.0±2.0	0.0±0.0	0.000±0.000	0.297	–

A: Diffusion — Seed 6

Method	SR	CS	G&C;	Steps±std	Viol#±std	Viol Σ ±std	Time	TE
dpcc-r	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.019	–
dpcc-r-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.948	–
dpcc-c	100%	0%	0%	84.5±11.5	19.0±17.0	0.381±0.377	0.704	–
dpcc-c-tight	100%	100%	100%	56.5±2.5	0.0±0.0	0.000±0.000	0.997	–
dpcc-t	100%	50%	50%	81.5±18.5	17.0±17.0	0.416±0.416	0.697	–
dpcc-t-tight	100%	100%	100%	62.5±2.5	0.0±0.0	0.000±0.000	0.964	–
diffuser	100%	0%	0%	59.0±3.0	7.5±0.5	0.090±0.001	0.307	0.025
gradient	100%	0%	0%	59.0±3.0	7.0±1.0	0.064±0.007	0.331	–
gradient-tight	100%	0%	0%	61.0±3.0	6.0±1.0	0.044±0.004	0.332	–
post_proc	100%	0%	0%	61.5±0.5	2.5±0.5	0.002±0.001	1.029	–
post_proc-tight	100%	100%	100%	66.0±0.0	0.0±0.0	0.000±0.000	0.924	–
model_free	100%	0%	0%	63.0±2.0	8.5±0.5	0.103±0.008	0.465	–
model_free-tight	100%	0%	0%	63.5±2.5	8.5±0.5	0.091±0.010	0.481	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5±0.5	0.5±0.5	0.000±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5±4.5	0.0±0.0	0.000±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5±2.5	0.0±0.0	0.000±0.000	0.737	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.0±2.0	0.0±0.0	0.000±0.000	0.691	–

Section 3 — Three-Way Head-to-Head Comparison

Best G&C; highlighted green. Best (lowest) violation count highlighted teal. ΔHG = H minus G G&C; score — the isolated RK4 effect. G Time and H Time columns allow direct per-method solver overhead comparison.

Top-Right-Hard

Method	A G&C;	G G&C;	H G&C;	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
dpcc-r	100%	50%	50%	A	=	0.0	4.5	4.5	0.275	0.261
dpcc-r-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.318	0.324
dpcc-c	100%	0%	0%	A	=	0.0	1.5	1.5	0.253	0.252
dpcc-c-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.349	0.360
dpcc-t	50%	0%	0%	A	=	0.5	2.5	2.5	0.275	0.294
dpcc-t-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.314	0.343
diffuser	100%	0%	0%	A	=	0.0	23.5	23.5	0.175	0.189
gradient	100%	50%	50%	A	=	0.0	16.0	16.0	0.175	0.201
gradient-tight	0%	50%	50%	G/H	=	15.0	16.0	16.0	0.180	0.208
post_proc	100%	50%	50%	A	=	0.0	4.5	4.5	0.265	0.304
post_proc-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.321	0.365
model_free	0%	0%	0%	A/G/ H	=	3.0	24.5	24.5	0.224	0.252
model_free-tight	50%	0%	0%	A	=	2.0	24.5	24.5	0.232	0.262
dpcc-c-tight-dt0.25	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.681	0.769
dpcc-c-tight-dt0.5	0%	100%	100%	G/H	=	1.0	0.0	0.0	0.426	0.489
dpcc-c-tight-dt2.0	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.272	0.330
dpcc-c-tight-dt4.0	50%	100%	100%	G/H	=	0.0	0.0	0.0	0.270	0.290

Top-Left-Hard

Method	A G&C;	G G&C;	H G&C;	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
dpcc-r	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.262	0.290
dpcc-r-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.346	0.396
dpcc-c	50%	100%	100%	G/H	=	1.0	0.0	0.0	0.258	0.301
dpcc-c-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.282	0.320
dpcc-t	50%	50%	50%	A/G/ H	=	1.5	22.0	22.0	0.257	0.287

dpcc-t-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.259	0.300
diffuser	0%	0%	0%	A/G/ H	=	20.0	24.5	24.5	0.173	0.195
gradient	0%	0%	0%	A/G/ H	=	17.5	16.0	16.0	0.185	0.211
gradient-tight	0%	0%	0%	A/G/ H	=	19.5	16.0	16.0	0.185	0.209
post_proc	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.254	0.296
post_proc-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.347	0.399
model_free	0%	0%	0%	A/G/ H	=	19.0	25.5	25.5	0.234	0.267
model_free-tight	0%	0%	0%	A/G/ H	=	15.0	25.5	25.5	0.239	0.275
dpcc-c-tight-dt0.25	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.529	0.594
dpcc-c-tight-dt0.5	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.329	0.366
dpcc-c-tight-dt2.0	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.268	0.287
dpcc-c-tight-dt4.0	0%	100%	100%	G/H	=	16.5	0.0	0.0	0.254	0.285

Both-Hard

Method	A G&C;	G G&C;	H G&C;	Best	ΔHG	A Viol	G Viol	H Viol	G Time	H Time
dpcc-r	0%	50%	50%	G/H	=	2.5	35.5	1.0	0.331	0.504
dpcc-r-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.337	0.413
dpcc-c	0%	0%	0%	A/G/ H	=	19.0	55.5	0.5	0.286	0.474
dpcc-c-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.519	0.633
dpcc-t	50%	0%	0%	A	=	17.0	65.5	0.0	0.294	0.510
dpcc-t-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.398	0.475
diffuser	0%	50%	50%	G/H	=	7.5	2.0	2.0	0.173	0.209
gradient	0%	50%	50%	G/H	=	7.0	2.0	2.0	0.187	0.219
gradient-tight	0%	0%	0%	A/G/ H	=	6.0	8.5	1.5	0.186	0.217
post_proc	0%	50%	50%	G/H	=	2.5	35.5	1.0	0.335	0.478
post_proc-tight	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.331	0.415
model_free	0%	50%	50%	G/H	=	8.5	2.0	2.0	0.239	0.291
model_free-tight	0%	50%	50%	G/H	=	8.5	1.5	1.5	0.250	0.305

dpcc-c-tight-dt0.25	50%	100%	0%	G	-100p p	0.5	0.0	0.0	0.666	0.969
dpcc-c-tight-dt0.5	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.449	0.554
dpcc-c-tight-dt2.0	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.323	0.401
dpcc-c-tight-dt4.0	100%	100%	100%	A/G/ H	=	0.0	0.0	0.0	0.297	0.383

Section 4 — dt Sensitivity: Complete Data

Note the dt=0.25 regression on both-hard: G=100%, H=0%. Also noteworthy: RK4 substantially raises compute time at small dt (both-hard dt=0.25: G=0.666s/step → H=0.969s/step).

Top-Right-Hard

dt	A G&C;	G G&C;	H G&C;	ΔHG	G Steps±std	H Steps±std	A Time	G Time	H Time
dt=0.25	100%	100%	100%	=	78.5±3.5	78.5±3.5	2.225	0.681	0.769
dt=0.5	0%	100%	100%	=	77.5±5.5	77.5±5.5	1.668	0.426	0.489
dt=1.0 (def)	100%	100%	100%	=	71.0±5.0	71.0±5.0	0.911	0.349	0.360
dt=2.0	100%	100%	100%	=	71.0±6.0	71.0±6.0	0.689	0.272	0.330
dt=4.0	50%	100%	100%	=	131.5±1.5	131.5±1.5	0.535	0.270	0.290

Top-Left-Hard

dt	A G&C;	G G&C;	H G&C;	ΔHG	G Steps±std	H Steps±std	A Time	G Time	H Time
dt=0.25	100%	100%	100%	=	76.0±2.0	76.0±2.0	1.259	0.529	0.594
dt=0.5	100%	100%	100%	=	68.5±2.5	68.5±2.5	0.937	0.329	0.366
dt=1.0 (def)	100%	100%	100%	=	67.0±2.0	67.0±2.0	0.822	0.282	0.320
dt=2.0	100%	100%	100%	=	67.0±3.0	67.0±3.0	0.679	0.268	0.287
dt=4.0	0%	100%	100%	=	124.5±4.5	124.5±4.5	0.810	0.254	0.285

Both-Hard

dt	A G&C;	G G&C;	H G&C;	ΔHG	G Steps±std	H Steps±std	A Time	G Time	H Time
dt=0.25	50%	100%	0%	-100p p	70.0±2.0	0.0±0.0	1.835	0.666	0.969
dt=0.5	100%	100%	100%	=	62.0±5.0	62.0±5.0	1.185	0.449	0.554
dt=1.0 (def)	100%	100%	100%	=	58.5±3.5	58.5±3.5	0.997	0.519	0.633
dt=2.0	100%	100%	100%	=	72.5±8.5	72.5±8.5	0.737	0.323	0.401
dt=4.0	100%	100%	100%	=	101.0±2.0	101.0±2.0	0.691	0.297	0.383

Section 5 — Summary Statistics & ODE Solver Analysis

5.1 Aggregate Statistics — Core Methods (13×3=39 observations each)

Metric	A (Diffusion)	G (Euler)	H (RK4)	H-G Δ	H-A Δ
G&C; Success	0.551	0.526	0.526	+0.000	-0.026
Constraint Sat.	0.551	0.526	0.526	+0.000	-0.026
Task Success	0.987	0.974	0.885	-0.090	-0.103
Avg Violations	4.936	11.667	6.628	-5.038	+1.692
Avg Total Viol.	0.372	0.722	0.624	-0.099	+0.252
Avg Time (s/step)	0.671	0.269	0.321	+0.051	-0.350

5.2 Win Counts — G&C; (39 comparisons)

A: 8	G (Euler): 0	H (RK4): 0	Ties: 31
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5.3 Key Findings: RK4 vs Euler (H vs G)

1. G&C; quality is nearly identical — RK4 provides no systematic improvement

G and H produce identical G&C; scores on 39/39 method×scenario pairs (100%). The trajectories sampled by the FMv3 continuous-t velocity field are smooth enough that Euler and RK4 converge to the same action sequence. Higher-order integration does not help when the velocity field is already well-conditioned.

2. RK4 reintroduces the dt=0.25 failure on both-hard — a clear regression

G (Euler) solved the dt=0.25 instability on both-hard (100% G&C; , 70 steps, 0 violations). H (RK4) fails at 0% G&C; on the same setting. RK4's larger effective step at each sub-evaluation appears to amplify the trajectory divergence at very small dt in the dual-obstacle scenario. This is a meaningful regression for the most challenging configuration.

3. RK4 improves both-hard dpcc-r and dpcc-t in some respects

H's dpcc-r on both-hard achieves 50% G&C; (60 steps, 1 violation) vs G's 50% (92 steps, 35.5 violations). H's dpcc-t on both-hard scores 0% but cleanly (0 steps, 0 violations) vs G's 0% with 135 steps and 65.5 violations — RK4 exits earlier rather than thrashing. Similarly H's dpcc-c on both-hard fails cleanly (0 steps) vs G's catastrophic 138 steps / 55.5 violations.

4. RK4 overhead is real but modest — roughly 19% average slowdown

H averages 0.321s/step vs G's 0.269s/step across core methods. The theoretical 4× ceiling is not reached because the DPCC projection dominates wall-time. The overhead is most visible on DPCC methods at small dt: dpcc-c-tight-dt0.25 on both-hard goes from 0.666s (G) to 0.969s (H) — a 46% increase. Baseline methods (diffuser, gradient) see ~15–25% overhead.

5. Top-right-hard and top-left-hard results are fully identical between G and H

Every method on top-right-hard and top-left-hard produces the same G&C; score, violation count, step count, and violation magnitude for G and H. Only both-hard shows any differences, suggesting the dual-obstacle interaction is the only scenario sensitive to ODE solver order at 10 steps.

6. Recommendation: Euler (G) remains the preferred solver

G wins 0 comparisons, H wins 0 (ties=31). H introduces a dt=0.25 regression on both-hard while providing no quality improvement on any other configuration. The marginal overhead from RK4 is not justified. G (FMv3 Euler, ODE=10) with tightened DPCC methods at dt=0.5–2.0 remains the recommended deployment configuration across all scenarios.

A=Diffusion (diffusion/6, Seed 6). G=FMv3 Euler (flow_matching_v3/6, continuous-t, explicit Euler, ODE steps=10, ckpt 90000). H=FMv3 RK4 (flow_matching_v3_ode_selectable/6, continuous-t, RK4, ODE steps=10, ckpt 90000). All Seed 6, avoiding-d3il, H8_K20_Dmodels.